

Technology News

Light switch for the brain

Ed Boyden, a neuroscientist at the Massachusetts Institute of Technology, has developed a way to silence parts of a brain just by shining light on them. The brain regions can be reactivated instantaneously by simply turning off the light. Boyden's discovery is a powerful tool for neuroscientists struggling to understand the complexity of the brain. With it, researchers will be able to probe how the circuitry of the brain works by silencing certain areas or types of brain cells and studying the effects. While early work will be done in animals, Boyden's discovery could soon be used as a prosthetic device in humans. It could quickly, and temporarily, shut down overactive brain regions implicated in conditions like epilepsy and depression.

Boyden analysed databases of genes in nature that help produce proteins that react to light, and found a pair in bacteria and fungi. Using

what appeared to be a harmless virus, he delivered a gene into specific brain cells. The gene instructed the cells to make pumps that pumped protons out of the cell when hit with a certain colour of light. This changed the electrical charge of the cell, preventing it from sending signals.

Flying robot to rescue people

Researchers at Chiba University, Japan, have developed a flying robot that flaps its wings 30 times a second, just like a real hummingbird.

Hiroshi Ryu, one of the researchers at the university, said the hummingbird robot is equipped with a micro motor and four wings, which enable it to fly freely in mid-air with rapid wing movements. The 2.6gm robot is controlled with an infrared sensor and can turn up, down, right or left. Developed at a cost of around \$2.1 million, it could be used to rescue people trapped in destroyed buildings, search for criminals

or even probe on Mars.

The researcher also plans to equip the robot with a mini camera by March 2011.

New alert system for cars

A new warning system for cars that alerts the driver if he gets distracted or is driving in drowsiness, has been developed by Spanish researchers.

Professor Josi Marma Armingol, Department of Systems Engineering and Automation of the Universidad Carlos III de Madrid, says, "The information comes to the system through a video camera, located on the dashboard of the car, that measures eye blinking speed, its degree of openness or where the driver is looking in order to detect potentially dangerous situations."

This system is named as Advanced Driving Alert System (ADAS) and can be fitted on any car.

Slimmest electronic reader with e-paper display

Skiff has rolled out an electronic reader with 29cm full-touchscreen display. The company claims 'Skiff Reader' has the largest and highest-resolution (1200x1600-pixel) electronic-paper display announced to date.

Also claimed to be the thinnest e-reader, Skiff Reader is optimised for newspaper and magazine content. It enables users to navigate and engage with the newspapers, magazines, books and other digital content they purchase through the Skiff Store.

This is the first device to feature e-paper technology from LG Display. The non-glass display avoids breakage risk to the device.

New display technology for TV sets

San Jose-based Prysm has come up with a laser phosphor display tech-



Professor Hiroshi Ryu of Japan's Chiba University displays his flying robot, which flaps its wings 30 times per second like a hummingbird, at his laboratory in Chiba city, suburban Tokyo